

D.A.V. LOHAGHAT
PRE-BOARD EXAMINATION-2023-24

SUBJECT - PHYSICS

MM: 70

CLASS-XII

Time: 3:00 hours

General Instructions :

Read the following instructions very carefully and strictly follow them :

1. This question paper contains **33** questions. **All** questions are **compulsory**.
2. This question paper is divided into **five** Sections **A, B, C, D** and **E**.
3. In **Section A** Questions no. **1** to **16** are Multiple Choice (MCQ) type questions, carrying **1** mark each.
4. In **Section B** Questions no. **17** to **21** are Very Short Answer (VSA) type questions, carrying **2** marks each.
5. In **Section C** Questions no. **22** to **28** are Short Answer (SA) type questions, carrying **3** marks each.
6. In **Section D** Questions no. **29** to **31** are Long Answer (LA) type questions carrying **5** marks each.
7. In **Section E** Questions no. **32** and **33** are case-based questions carrying **4** marks each.
8. There is no overall choice. However, an internal choice has been provided in 1 questions in Section B, 1 questions in Section C, 3 questions in Section D and one question in each CBQ in Section E. You have to attempt only one of the choices in such questions.
9. Use of calculators is **not** allowed.
10. Use the following values of physical constants, if required :
 $c = 3 \times 10^8 \text{ m/s}$
 $h = 6.63 \times 10^{-34} \text{ Js}$
 $e = 1.6 \times 10^{-19} \text{ C}$
 $\mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1}$
 $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$
 $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2\text{C}^{-2}$
Mass of electron (m_e) = $9.1 \times 10^{-31} \text{ kg}$
Mass of neutron = $1.675 \times 10^{-27} \text{ kg}$
Mass of proton = $1.673 \times 10^{-27} \text{ kg}$
Avogadro's number = 6.023×10^{23} per gram mole
Boltzmann constant = $1.38 \times 10^{-23} \text{ JK}^{-1}$

Section A - MCQ

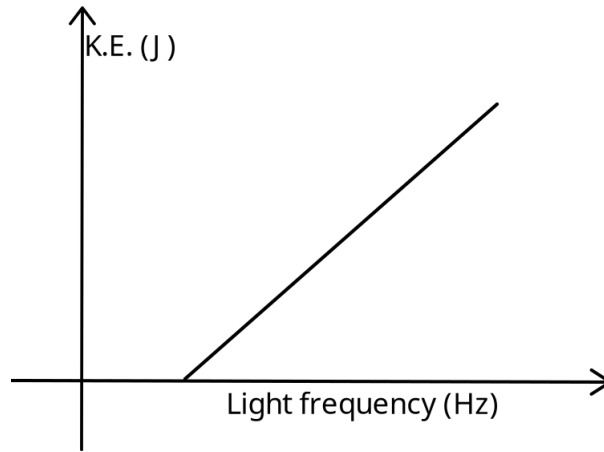
Q1. An electric dipole having a dipole moment of $4 \times 10^{-9} \text{ Cm}$ is placed in a uniform electric field such that the dipole is in stable equilibrium. If the magnitude of the electric field is $3 \times 10^3 \text{ N/C}$, what is the work done in rotating the dipole to a position of unstable equilibrium? [1]

- A. Zero B. $1.2 \times 10^{-5} \text{ J}$ C. $2.4 \times 10^{-5} \text{ J}$ D. $-1.2 \times 10^{-5} \text{ J}$

Q2. An infinite line of charge has a linear charge density of 10^{-7} C/m . What will be the magnitude of the force acting on an alpha particle placed at a distance of 4 cm from the line of charge? [1]

- A. $14.4 \times 10^{-15} \text{ N}$ B. $7.2 \times 10^{-15} \text{ N}$ C. $4.5 \times 10^4 \text{ N}$ D. $9 \times 10^4 \text{ N}$

Q3. The graph below shows the variation of the maximum kinetic energy of the emitted photo-electron with the frequency of the incident radiation for a given metal. [1]



Which of the following gives the work function of the metal?

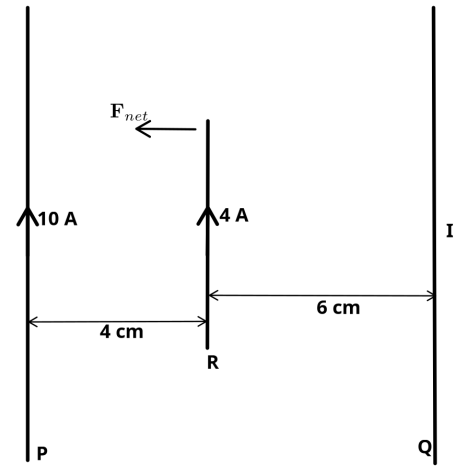
- A. x-intercept C. the slope of the graph
B. y-intercept D. the area under the graph

Q4. When an electron in an atom moves from the ground state to a higher energy level what happens to its kinetic and potential energies? [1]

	Kinetic energy	Potential energy
A	increases	increases
B	increases	decreases
C	decreases	increases
D	decreases	decreases

Q 5. Two long and straight current-carrying wires, **P** and **Q** are placed parallel to each other separated by a distance of **10 cm**. A wire '**R**' of length **8 cm** and carrying a current of **4 A** is placed between the two wires **P** and **Q** as shown below.

If the wire **R**, experiences a net force towards wire **P**, then which of the following is **definitely TRUE** about the current '**I**' in wire **Q**?

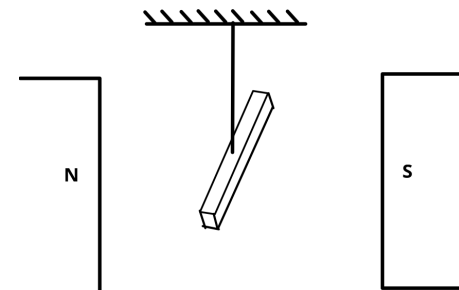


[1]

- A. Current **I** cannot be in the upward direction.
- B. Current **I** can have any magnitude greater than **0 A** in the upward direction.
- C. Current **I** cannot have a magnitude of more than **15 A** in the upward direction.
- D. Current **I** cannot have a magnitude of more than **10 A** in the upward direction.

Q 6. A rod when suspended in a uniform magnetic field aligns itself perpendicular to the magnetic field as shown below.

Which of the following statements is/are true for the rod?



[1]

- P) Every atom in the rod, has a zero magnetic moment in the absence of external field.
- Q) The rod is attracted when taken near the poles of a strong magnet.
- R) The relative permeability of the material of the rod is slightly less than 1.
- S) The susceptibility of the material of the rod is directly proportional to temperature.

- A. only **Q**
- B. only **P** and **R**
- C. only **Q** and **S**
- D. only **R** and **S**

Q 7. Three students construct a solenoid of length **35 cm**. They are each given insulated copper wire of the same length. The table below lists some details about the solenoids made by them.

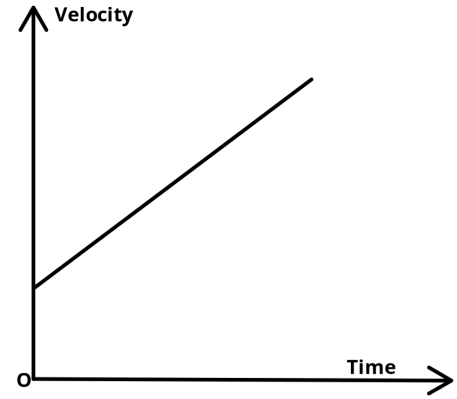
[1]

	Magnetic field produced	Radius of solenoid	Core of solenoid
Student 1	B_1	3 cm	air
Student 2	B_2	3 cm	iron
Student 3	B_3	6 cm	air

Compare the magnetic field produced by the solenoids made by the three students.

- A. $B_1 = B_3 < B_2$ B. $B_3 < B_1 < B_2$ C. $B_1 < B_2 < B_3$ D. $B_1 = B_2 > B_3$

Q8. A charged particle '+q' having a mass 'm' moves in a uniform electric and magnetic field. In which of the following scenarios will the path of the charged particle be linear and described by the velocity time graph shown below?

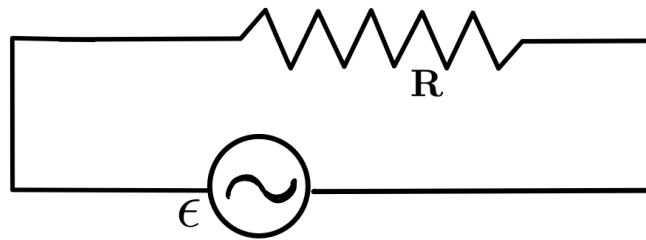


[1]

- A. $E \perp B \perp$ velocity of the particle
 B. $E \parallel B$ and the particle is initially at rest
 C. $E \parallel B$ and the particle has an initial velocity along the electric field
 D. $E \perp B$ and the particle has an initial velocity along the electric field

Q9. A pure resistor is connected to an AC power source as shown below.

[1]



Which of the following statement(s) is/are TRUE?

- I: The average current flowing through the circuit during one full cycle is zero.
 II: The current in the resistor leads the voltage by $\frac{\pi}{2}$.
 III: The average power dissipated by the resistor is zero.

- A. only I B. only I and II C. only II and III D. all- I, II and II

Q10. At what rate does the electric field change between the plates of a square capacitor of side 5 cm, if the plates are spaced 1.2 mm apart and the voltage across them is changing at a rate of 60 V/s?

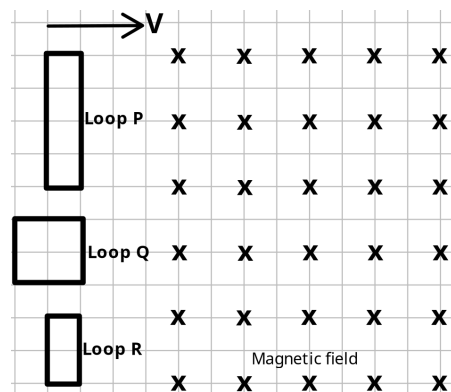
[1]

- A. $7.2 \times 10^{-2} \text{ Vm}^{-1}\text{s}^{-1}$ C. $12 \times 10^2 \text{ Vm}^{-1}\text{s}^{-1}$
 B. $30 \times 10^{-1} \text{ Vm}^{-1}\text{s}^{-1}$ D. $5 \times 10^4 \text{ Vm}^{-1}\text{s}^{-1}$

Q 11. Three loops as shown in the figure move into the magnetic field with a velocity v .

In which loop(s) will the induced emf be the **largest** at the instant when the loops enter the magnetic field?

- A. Only P
- B. Only Q
- C. Only P and Q
- D. Only Q and R

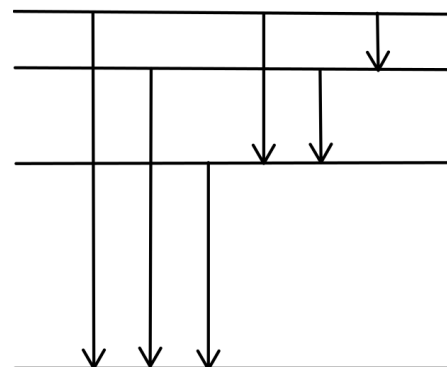


[1]

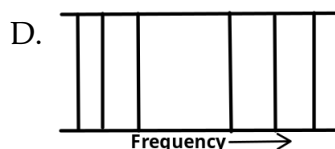
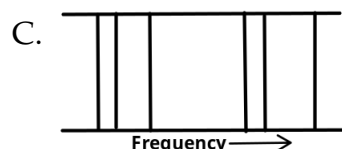
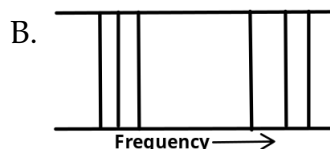
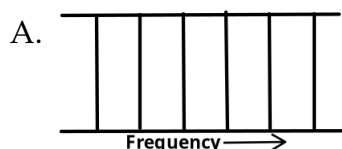
Q 12. The emission spectrum of an element is the spectrum of frequencies of em radiations emitted due to electrons making a transition from a higher energy state to a lower energy state.

The diagram shows electrons transitioning from higher energy states to lower energy states.

Which of the following spectrums most closely corresponds to the above transitions?



[1]



Questions number 13 to 16 are Assertion (A) and Reason (R) type questions. Two statements are given - one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (a), (b), (c) and (d) as given below.

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
- (b) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).
- (c) Assertion (A) is true, but Reason (R) is false.
- (d) Assertion (A) is false and Reason (R) is also false.

Q 13. Assertion (A) : The work function of a given material increases with an increase in the frequency of the incident radiation.

[1]

Reason (R) : As per Einstein's photoelectric equation $h\nu = \phi + K.E$, work function ϕ is directly proportional to the frequency ν of the incident radiation.

Q 14. Assertion (A) : The conductivity of intrinsic semiconductors increases with an increase in temperature.

[1]

Reason (R) : Increase in temperature decreases the average time between collisions of electrons.

Q 15. Assertion (A) : The direction of the electric field is always perpendicular to the equipotential surface. [1]

Reason (R) : Work is done by the electric force in moving a charge between any two points on an equipotential surface is zero.

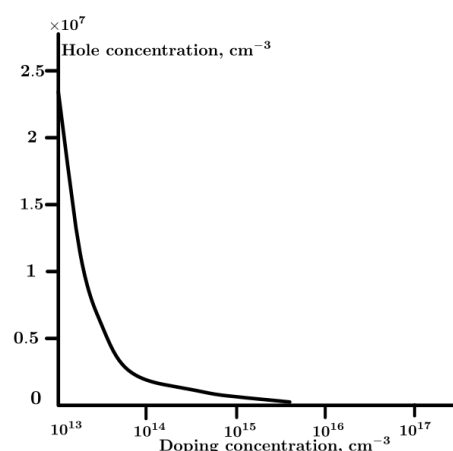
Q 16. Assertion (A) : If the focal length of two convex lenses is the same, the lens with the larger diameter will produce brighter images. [1]

Reason (R) : Convex lenses with larger diameters are able to focus light better.

Section B - Very Short Answer Questions

Q 17. The graph shows the variation in hole concentration with doping concentration in an extrinsic semiconductor doped with pentavalent impurities.

Why does the hole concentration reduce when pentavalent doping is increased?



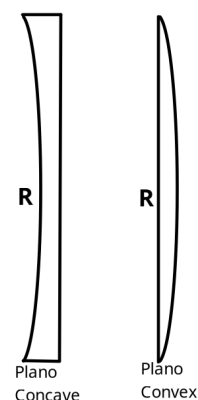
Q 18. λ_α and λ_p are the wavelengths associated with a moving alpha particle and a proton respectively. [2]

Obtain the relation between velocities of the two particles for which,

i. $\lambda_\alpha > \lambda_p$

ii. $\lambda_\alpha = \lambda_p$

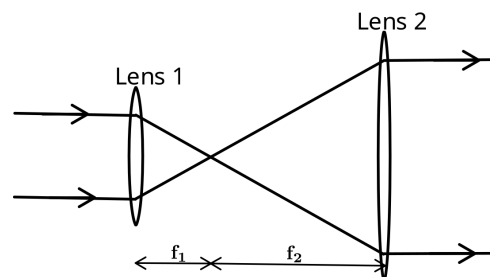
Q 19. Compare the focal lengths of the two lenses shown below if the radius of curvature of the curved surface is the same in both lenses. [2]



Q 20. Two copper wires, P and Q of the same area of cross-section are joined in parallel. The combination of wires is connected across a battery of potential difference V . If the length of the wires, P and Q are in the ratio 1:2, find the ratio of drift velocities of electrons in wires P and Q. [2]

- Q 21.** (a) The image below shows a setup of a device that is used to increase the diameter of a light beam from a laser.

Show how a combination of a convex and a concave lens can also be arranged to increase the diameter of a light beam. Your answer should include how the two lenses should be arranged and the distance between the two lenses. (Note that the rays in both the incident and emergent beam are parallel.)

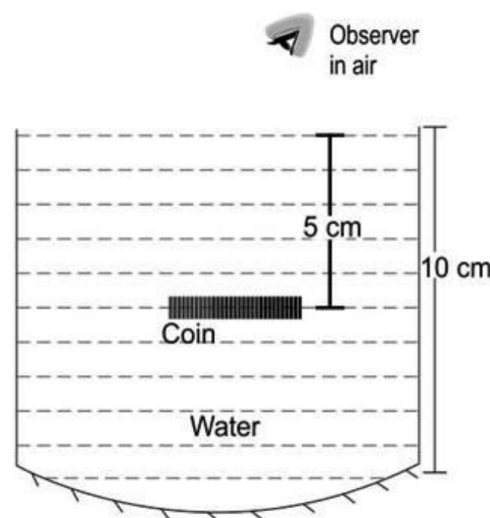


[2]

OR

- (b) A glass beaker of height 10 cm, completely filled with water (refractive index = $\frac{4}{3}$), has a curved bottom which is silvered as shown below.

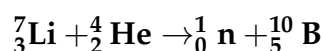
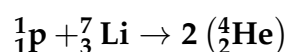
A plastic coin remains submerged in water at a depth of 5 cm from the top of the beaker. An observer sees the coin in the water and its image in the mirror. If the image formed by the curved mirror is seen by the observer at a distance of 15 cm from the surface of the water, what is the focal length of the curved surface? (Assume the silvered curved surface acts as a spherical mirror.)



[2]

Section C - Short Answer Questions

- Q 22.** Identify if the two nuclear reactions mentioned below are endothermic or exothermic. Show your calculations.



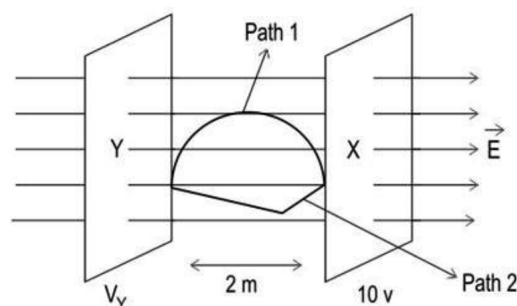
Use the information below to answer the question:

$${}_1^1\text{p} = 1.00728 \text{ amu}, \quad {}_3^7\text{Li} = 7.0160 \text{ amu}, \quad {}_2^4\text{He} = 4.0026 \text{ amu}$$

$${}_0^1\text{n} = 1.0087 \text{ amu}, \quad {}_5^{10}\text{B} = 10.01294 \text{ amu}$$

- Q 23.** X and Y are two equipotential surfaces separated by a distance of 2 m in a uniform electric field of 10 V/m. Surface X has a potential of 10 V

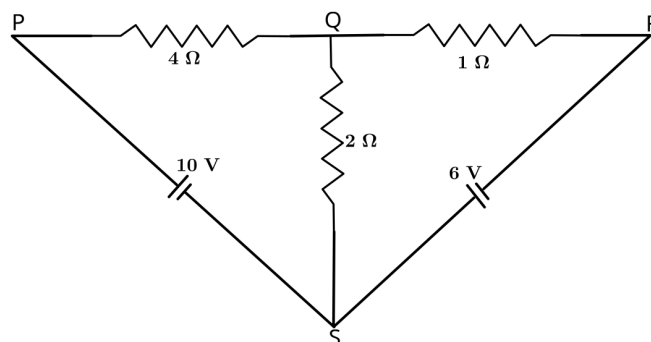
- (a) Calculate the potential of surface Y.
 (b) What is the work done in moving a +2 C charge from surface Y to surface X along path 1? How will this work change when the charge is moved along Path 2? Give a reason for your answer.



[3]

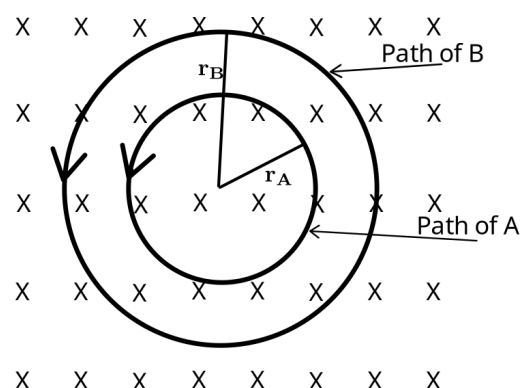
- Q 24.** (a) Compare the de Broglie wavelength associated with the electron in the third orbit to the circumference of the orbit. [3]
- (b) In which of the following will the electrons have the same de Broglie wavelength?
- Third orbit of He atom
 - Fourth orbit of He atom
 - Third orbit of Li atom
 - Sixth orbit of Be atom
- Show your calculations.

- Q 25.** Using Kirchhoff's laws, calculate the current flowing through $4\ \Omega$, $1\ \Omega$ and $2\ \Omega$ resistors in the circuit shown below. [3]



- Q 26.** Two charges **A** and **B**, each having a velocity of $\mathbf{v}$, traverse circular paths in a uniform magnetic field as shown below.

- (a) Compare the charge-to-mass ratio of the two particles A and B. Show the necessary mathematical calculations.
- (b) Which of the two particles is likely to be a proton if the other is an alpha particle? Give reason.



- Q 27.** (a) A radio wave and an infrasonic wave have the same wavelength when travelling through air. Are their frequencies the same or different? Give a reason for your answer. [3]
- (b) An electromagnetic wave traveling east has a magnetic field that oscillates vertically and has a frequency of **60 kHz** and an rms strength of $8 \times 10^{-9}\ \text{T}$. Determine the frequency and the rms strength of the electric field. What is the direction of the electric field?
- Q 28.** (a) A circular ring of diameter 0.2 m is placed in a uniform magnetic field of **0.4 T**. The ring is rotated about its diameter at a frequency of **60 Hz**. [3]
- If the ring has 50 turns, then what is the maximum induced emf in the ring?
 - State one condition under which the induced emf in the circular ring will be zero?

OR

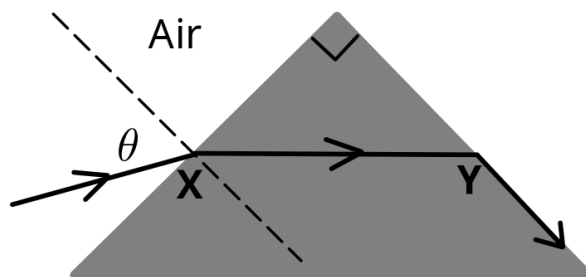
- (b) Given below are a few characteristics of solenoids **p** and **q**.

	solenoid p	solenoid q
length of the solenoid	1 (m)	1 (m)
number of turns (N)	200	50
cross-sectional area of the wire	$A \text{ (m}^2\text{)}$	$A \text{ (m}^2\text{)}$
relative permeability of the core material	1	500
self-inductance	2 (mH)	?

What is the self-inductance of the solenoid q?

Section D - Long Answer Questions

- Q 29.** (a) i. A ray of light is incident at angle θ on a right-angled prism at point X. At point Y, it emerges along the prism surface. Calculate the refractive index of the prism in terms of the incident angle. [5]



- ii. Show that for an equilateral prism kept in air, minimum deviation occurs when the angle of incidence $i = \sin^{-1}\left(\frac{n}{2}\right)$, where n is the refractive index of the material of the prism.

OR

- (b) i. A Young's double slit setup is illuminated with monochromatic light. If the intensity of light passing through one of the slits is reduced, explain the changes that can be seen in the appearance of the bright and dark fringes? [5]
- ii. α) A single slit diffraction setup is illuminated with green light of wavelength **500 nm**. If the width of the slit is **1 mm** and the screen is **2 m** away from the slits, calculate the width of the central maximum.
- β) What will happen to the width of the central maximum, if the green light is replaced with the red light? Give a reason for your answer.
- iii. A student wishes to study the diffraction of sound using the single slit setup. He replaces the light source with a sound source. What other change should he do to study the diffraction pattern?

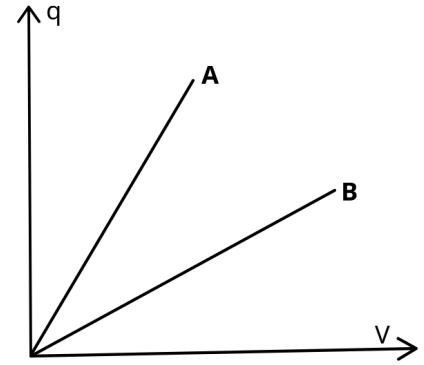
- Q 30.** (a) i. A camera usually operates at **1.5 V** and this potential difference is not sufficient to emit light energy using flash. For this purpose, the flash circuit of the camera has a capacitor that is charged to **300 V-330 V** using various electrical components. If the voltage generated across the plates of the capacitor is **300 V** and the capacitance of the parallel plate capacitor used is **100 μF** , then find the energy released when the trigger button on the camera is pressed. [5]
- ii. How much charge does the **100 μF** capacitor charged to **300 V** hold?

- iii. If the distance between the parallel plate capacitor of capacitance $100 \mu F$ is increased two times, then calculate the capacitance of the capacitor.
- iv. The graph below shows the variation of charge ' q ' with potential difference ' V ' for a parallel plate capacitor ' C ' for scenarios P and Q.

Scenario P - the space between the capacitor ' C ' is filled with **air**.

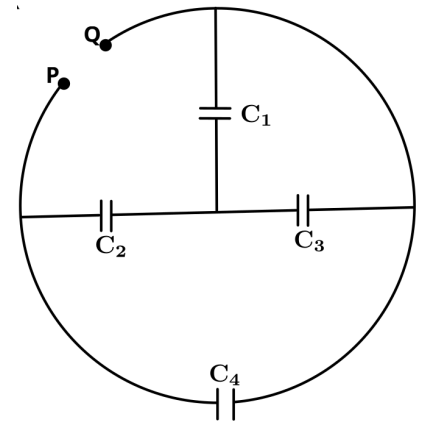
Scenario Q - the space between the capacitor ' C ' is filled with a substance of **dielectric constant K**.

Which of the two lines **A** or **B** corresponds to scenario **Q**? Give a reason for your answer.



OR

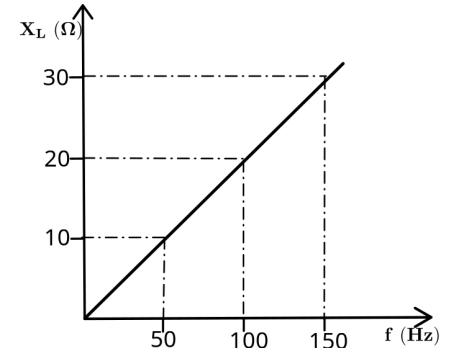
- (b) i. Find the effective capacitance between points **P** and **Q**, if each capacitor has a capacitance of $6 \mu F$
- ii. Find the ratio of charges on capacitors C_1 and C_4 , if the potential difference between points **P** and **Q** is **10 V**.



[5]

- Q31.** (a) An inductor of inductance ' L ' is connected to an AC source, $V = 100 \sin \omega t$.

The graph below represents the variation of inductive reactance X_L of the inductor with the frequency of an alternating source.



[5]

- i. What is the self-inductance of the inductor?
- ii. If the ac source is replaced by a battery such that $V = 100 V$, then what is the inductive reactance of the inductor? Give reason.
- iii. When the frequency is **50 Hz**, what is the average power dissipated by the inductor over a complete cycle in the circuit? Justify your answer.
- iv. This inductor is connected in series with a resistance of 15Ω and a capacitor of $5 \mu F$. The frequency of the alternating source is varied such that the power dissipated in the circuit becomes maximum. Calculate the frequency and the phase difference between alternating voltage and current when the power dissipated is the maximum.

OR

- (b) An ideal transformer having a ferromagnetic core consists of two coils having **500 turns** (primary) and **50 turns** (secondary) respectively.

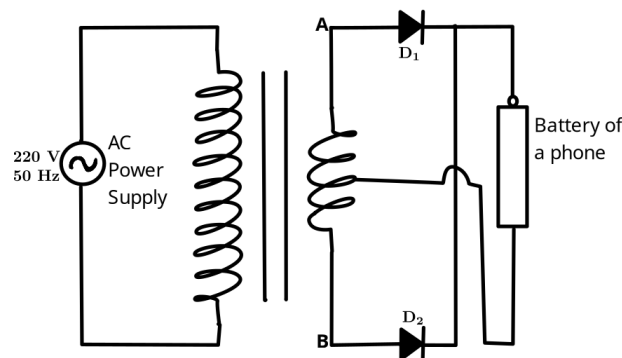
[5]

- What is the voltage across the secondary coil, if the rms voltage across the primary coil is **240 V**?
- What will be the individual currents in the two coils (primary and secondary), if the secondary has a resistive load of **20 Ω** ?

Section E - Case Study

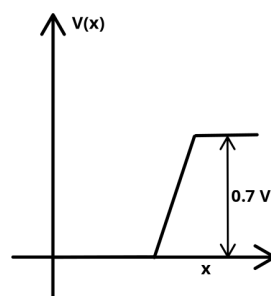
Q32. Read the following paragraph and answer the questions that follow. For most mobile devices, the voltage to recharge the battery is typically 5 volts of direct current. In India, the current supplied to our homes is alternating current at 220 V and at a frequency of 50 Hz. Fatima designed a simplified version of a mobile phone charger. She made a circuit using a centre tap transformer and two similar silicon diodes D_1 and D_2 as shown below. Study the diagram below and answer the questions that follow.

[4]



Answer the following questions based on the above:

- Can Fatima also charge the battery of a phone by connecting the battery directly to the ac power supply? Give reason.
- The graph of the potential barrier (V) vs width of the depletion region (x), when D_1 is unbiased at room temperature, is shown below.



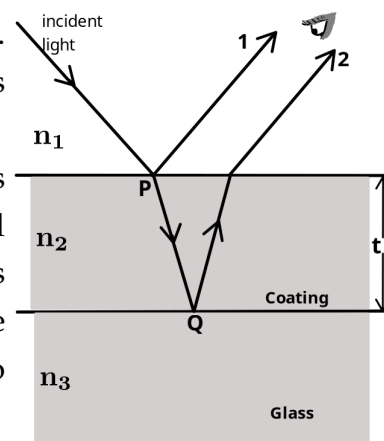
Plot a comparative graph of the potential barrier (V) vs width of the depletion region (x) of D_1 at room temperature when the voltage at A is negative with respect to voltage at centre tap. Give reason.

OR

- If the battery of the phone is directly connected to the output terminals of the secondary coil of the transformer, will it get charged? Justify your answer.
- What will be the output frequency across the phone's battery when the orientation of D_2 is reversed in fig. 1 and the centre-tapped three-output transformer is replaced by a two-output step-down transformer? Justify your answer.

- Q 33.** When light rays fall on glass, about 4 % of the light gets reflected. To eliminate this reflection, the glass display cases in museums usually have an anti-reflective coating.

This works on the principle of interference. When light falls on the coated glass, the light gets reflected from the top and bottom surfaces of the coating and these two reflected light rays can interfere. To reduce reflection, the thickness and refractive index of the coating are adjusted such that the light rays undergo destructive interference.



[4]

Reflected light undergoes a 180° phase shift when it falls on a denser medium from a rarer medium and no phase shift when it falls on a rarer medium from a denser medium. (Note: The thickness of coating is much less than the glass.) To answer the questions below, consider a monochromatic light of wavelength λ incident on the coating of thickness t at a small angle of incidence and $n_1 < n_2 < n_3$. Also Consider $PQ \approx t$.

- (a) Which of the following occurs, if there is no coating on the glass?
- The object behind the case looks distorted.
 - The colours of the object behind the glass case appear dull.
 - A reflection of the objects in front of the glass case is seen on the case.
 - Multiple reflections of the object behind the glass case are seen on the case
- (b) What is the path difference between rays 1 and 2? (Consider $PQ \approx t$.)
- t
 - $2t$
 - λ
 - 2λ
- (c) For what minimum thickness of the coating, do the two rays 1 and 2 undergo destructive interference? (Remember the wavelength of the light ray changes as it moves from one media to another.)
- $n_2\lambda/2$
 - $n_2\lambda/4$
 - $\lambda/(2n_2)$
 - $\lambda/(4n_2)$

OR

- (c) For what minimum thickness of the coating, do the two rays 1 and 2 undergo constructive interference? (Remember the wavelength of the light ray changes as it moves from one media to another.)
- $n_2\lambda$
 - $n_2\lambda/2$
 - $\lambda/(n_2)$
 - $\lambda/(2n_2)$
- (d) If the material of the coating is changed such that $n_2 > n_3$, what will be the additional path difference compared to the path difference identified in question (b)?
- t
 - π
 - $\lambda/2$
 - There will be NO additional path difference.